

DEEP LEARNING AND SEMI-SUPERVISED TECHNIQUES FOR SEGMENTATION OF MICRO-TOMOGRAPHY IMAGES

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ABSTRACT

X-ray micro-tomography enables detailed pore analysis, but the scarcity of annotated slices limits the effectiveness of fully supervised segmentation models. This study evaluates a semi-supervised cross-teaching framework combining a U-Net and a Vision Transformer (ViT) to leverage both labelled and unlabelled data. Using 22 labelled and 172 unlabelled slices, the cross-teaching U-Net achieves a Dice score of 0.97, greatly outperforming its supervised baseline. The ViT, however, underperforms and provides limited benefit to the ensemble. Overall, the results show that semi-supervised learning can substantially improve segmentation performance in low-annotation settings.

1. INTRODUCTION

X-ray micro-tomography has become a key method for investigating material micro-structures, enabling non-destructive and high-resolution analysis of porosity, crack formation and deformation processes. The dataset used for this project contained 2D slices extracted from a 3D volume scanned at 22 nm resolution that captures pores whose morphology reflects how the material responds to heat treatment. These pores must be accurately segmented to support downstream analyses, including pore tracking, deformation characterization, and structural modelling.

Due to the high resolution and the large volume of data, only a limited number of slices can be manually annotated, making fully supervised deep learning models difficult to apply reliably. This limitations motivates the use of semi-supervised learning models, where both the unlabelled and labelled slices contribute to model optimization.

Convolutional neural networks, particularly the U-Net architecture [1], remain the dominant approaches for biomedical and microscopy segmentation due to their ability to capture fine spatial detail through encoder-decoder structures with skip connections. More recently, Vision Transformers (ViTs) [2] have shown strong performance by modelling long-range dependencies through self-attention, providing stronger global context modelling. These architectures offer

complementary strengths, suggesting potential benefits from combining them.

Building on these observations, this project investigates whether combining these two methods in a cross-teaching semi-supervised framework can leverage their complementary strengths. In addition to evaluating a standard semi-supervised U-Net and ViT, a framework will be introduced in which both models exchange high-confidence pseudo-labels during training. The aim is to determine whether this enhances the robustness and precision of the segmentation in a setting with limited annotations, which is typical in micro-CT analysis workflows.

2. RELATED WORK

Deep learning has become a central approach for image segmentation, especially in biomedical and microscopy domains. Convolutional architectures such as U-Net [1] and its larger variants, including 3D U-Net [3] and V-Net [4], have demonstrated strong performance due to their ability to recover fine spatial detail through encoder-decoder structures with skip connections. These models have also been applied successfully to porous media and micro-CT data, where pore morphology and contrast variations require robust extraction of local features [5].

Transformer-based architectures introduced an alternative modelling approach through self-attention, enabling the capture of long-range dependencies and global structural relationships. The Vision Transformer (ViT) [2] laid the foundation for transformer-based vision models, while hybrid CNN-Transformer designs such as TransUNet [6] and FE-SwinUp [7] demonstrated improved segmentation performance by combining local feature extraction with global context modelling. Recent transformer-based approaches have also been explored for X-ray tomography segmentation, further motivating their relevance for high-resolution scientific imaging tasks [8].

The scarcity of labelled micro-CT images has motivated a growing body of research on semi-supervised segmentation. Surveys such as [9, 10, 11] show approaches including pseudo-labelling, consistency regularization, and

self-ensembling, all of them seek to leverage unlabelled data when annotations are limited. Cross-teaching and co-training frameworks extend these ideas by allowing multiple models to exchange pseudo-labels, reducing confirmation bias and improving training stability. Luo et al. [12] and Yang et al. [13] demonstrate that combining CNN and Transformer architectures in semi-supervised cross-teaching frameworks leads to improved segmentation performance, providing a strong motivation for evaluating a combined U-Net and ViT approach for micro-CT pore segmentation.

3. METHOD

3.1. Data and Problem Formulation

The objective of this study is to segment pores and deformation structures in high-resolution tomography images. The dataset comprises 22 labelled 2D slices and 172 unlabelled slices. Given the limited quantity of annotated data, a semi-supervised learning strategy was adopted based on cross-teaching, in which two heterogeneous segmentation models exchange pseudo-labels generated from unlabelled samples. This approach allows both networks to leverage complementary inductive biases, local convolutional processing and global self-attention, to improve segmentation quality under sparse supervision.

3.2. Model Architecture

3.2.1. U-Net

The first network is a U-Net with a ResNet-34 encoder pre-trained on ImageNet, implemented using the segmentation models pytorch framework. The encoder is frozen during training so that optimization affects only the decoder and segmentation head. The model accepts 1-channel 512×512 inputs and outputs a 2-class probability map (background, pore). Its convolutional architecture is well suited to capture local structural patterns characteristic of porous materials.

3.2.2. Vision Transdormer (ViT) Segmentation Model

The second network is based on ViT-Base (patch16-224) from the timm library, operating on 3-channel 224×224 images. The original classification head is removed, and a custom decoder composed of multiple transposed convolution layers reconstructs a dense segmentation map. The ViT backbone provides global receptive fields and long-range dependencies, which complement the locality of the U-Net.

3.3. Semi-Supervised Cross-Teaching Framework

To exploit the unlabelled data, we employ a mutual pseudo-labelling strategy inspired by cross-teaching. In this framework, each model serves as a teacher for the other by generating pseudo-labels for unlabelled samples.

3.3.1. Supervised learning on labelled Data

For labelled batches, both networks are trained independently using the provided ground-truth masks. The training procedure consists of the following. It starts by forward propagation through each model. Then Resizing the masks to match the model-specific input resolution. After which it computes the Cross-Entropy loss:

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(p_{i,c})$$

Then it updates the Parameter via Adam (learning rate 1×10^{-4}).

3.3.2. Unlabelled Learning via Cross-Teaching

For unlabelled samples, neither model has access to ground-truth annotations. Instead, both models generate predictions, which are then used as pseudo-labels for the other network. This done by the following steps: Each model produces a segmentation prediction for the unlabelled input. Then the pixel-wise confidence is computed using the maximum softmax probability. Predictions exceeding a threshold of 0.9 are considered reliable. After this the Pseudo-labels are exchanged across the models. The ViT predictions are upsampled to 512×512 and used as pseudo-labels for the U-Net. While the U-Net predictions are downsampled to 224×224 and used as pseudo-labels for the ViT. Hard labels are obtained through argmax over class probabilities. When the predictions are sampled to the correct size then each model is trained on the pseudo-labels using a consistency loss and afterwards scaled by a consistency eight of 0.5:

$$\mathcal{L}_{cons} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C \hat{y}_{i,c}^{(t)} \log p_{i,c}^{(s)}$$

This mechanism encourages the models to converge toward mutually consistent segmentations and allows the unlabelled set to contribute to representation learning.

3.4. Ensemble Inference

During inference, segmentation quality is improved by combining predictions from both models. The probability maps produced by the U-Net (after resizing the ViT output) are averaged:

$$\hat{y}_{ensemble} = \frac{1}{2} \left(\text{softmax}(f_{UNet}(x)) + \text{softmax}(f_{ViT}(x)) \right)$$

The final segmentation mask is obtained by applying a class-wise argmax to the ensemble probability map. This strategy leverages the complementary strengths of convolutional and transformer-based representations.

4. EXPERIMENTAL DESIGN

This section describes the datasets, preprocessing pipeline, training configuration, and evaluation strategy used to assess the proposed semi-supervised cross-teaching framework combining a U-Net and Vision Transformer (ViT). All experiments were conducted using the implementation detailed in Section 3. For the experiment 2 datasets were used. One consisted of labelled data and one consisted of unlabelled data. The labelled dataset consists of paired volumetric microscopy images and corresponding binary pore masks. The images were 2D slices of 22m X-rays stored in .tif files as high-res grayscale microscopy. All labelled images and masks follow a consistent naming scheme and are sorted to preserve matching order. The unlabelled dataset is drawn from a separate directory containing only raw microscopy images. These images were the same type of files, slightly larger in dimensions. These images are used exclusively for cross-teaching and do not contain ground-truth labels.

4.1. Preprocessing

All images, labelled and unlabelled, undergo a consistent preprocessing pipeline. First, pixel intensities are normalized to the range $[0, 1]$ according to

$$I_{\text{norm}} = \frac{I - \min(I)}{\max(I) - \min(I) + 10^{-8}}.$$

The labelled masks are binarized into two classes: background (0) and pore (1). Since the U-Net and ViT models expect inputs of different dimensionalities, channels are prepared accordingly. Grayscale images are replicated along the channel axis to create three-channel inputs for the ViT, while the U-Net receives a single-channel representation obtained by averaging across channels when needed. Images are also resized to match the expected input resolution of each model: U-Net to 512×512 , single-channel input and ViT to 224×224 , three-channel input. All resizing uses bilinear interpolation for the intensity images and nearest-neighbour interpolation for masks.

Augmentation is then applied on the training images this to provide variation and more labelled images to train on. The augmentations are from torchvision transforms and consist of vertical and horizontal flips along with random rotations.

4.2. Training Split

During the supervised training, the images were split in a standard training-validation-test split, which was automatically applied from the sklearn library. The unlabelled or semi-supervised training did not follow this approach as there are no labels to validate or test, therefore small incremental training steps were only taken if the models had high-confidence predictions to be used as pseudo-labels.

4.3. Training Configuration

The two segmentation models were initialized as pre-trained models; the U-Net from the 'segmentation_models_pytorch' library and the ViT from the 'timm' library. Both models then used supervised transfer learning to fine tune the segmentation heads of the models on the porosity dataset. Next, training jointly optimizes the U-Net and the ViT model, each with its own resolution and inductive biases.

Both models are trained using the Adam optimizer with a standard learning rate of 1×10^{-4} , a batch size of 8, and 30 training epochs. Supervised learning on labelled images uses cross-entropy loss. For unlabelled images, a cross-teaching mechanism is applied in which each model generates pseudo-labels for the other via the argmax of its logits if the confidence of a model is greater than 90 percent (averaged over the image). The consistency loss is weighted by 0.5. The system automatically selects the most suitable hardware backend (MPS, CUDA, or CPU).

4.4. Cross-Teaching Protocol

A key component of the training process is the bidirectional teacher-student interaction. For each unlabelled batch, both models generate predictions that are resized to match the other model's resolution and then converted into hard pseudo-labels. The U-Net is downsampled and the ViT is upsampled, so they both fit the correct logits. This interaction encourages both models to converge toward mutually consistent predictions, leveraging the complementary strengths of convolutional and transformer-based architectures.

4.5. Evaluation Strategy

Inference is performed using an ensemble strategy that combines the probabilistic outputs of the two models. The ViT predictions are upsampled to the U-Net resolution, and the final probability map is computed as

$$P_{\text{ens}} = \frac{1}{2} (\text{softmax}(f_{\text{UNet}}) + \text{softmax}(f_{\text{ViT} \rightarrow 512})).$$

A final segmentation mask is obtained by applying the argmax operator to the ensemble output. Although the current implementation does not automatically compute evaluation metrics, it supports common segmentation metrics such as the Dice coefficient, Intersection over Union (IoU), and pixel accuracy, which can be computed offline.

4.6. Baselines

To contextualize the performance of the proposed method, relevant baselines include a supervised U-Net and ViT only trained on labelled data. The proposed semi-supervised cross-teaching method. the last one a fully supervised model trained on densely labelled data (upper bound).

These baselines provide meaningful comparisons for assessing the impact of leveraging unlabelled data through the cross-teaching framework.

5. RESULTS

Table 1 shows the segmentation performance of all evaluated models using the Dice coefficient, IoU, precision and recall. From this can be seen that the stand-alone U-Net achieves a moderate performance (Dice 0.49), indicating that 22 annotated slices are insufficient for stable supervised training. The stand-alone ViT performs considerably better (Dice 0.89), suggesting that transformer-based architectures can extract global structure from the tomography images even with limited labels. Both cross-teaching models outperform their supervised stand-alone models: the cross-teaching U-Net reaches a Dice of 0.97, while the cross-teaching ViT achieves the highest score of 0.99. The ensemble model also performs strongly, slightly below the ViT but still demonstrating high segmentation quality. These results indicate that the cross-teaching framework improves both architectures and is particularly effective for the ViT.

Complete Model Comparison on X-ray Micro-CT Segmentation				
Model	Dice	IoU	Precision	Recall
ViT (Cross-Teach)	0.991645	0.983430	0.993447	0.989850
Ensemble (Cross-Teach)	0.988273	0.976825	0.982228	0.994413
U-Net (Cross-Teach)	0.969721	0.941409	0.948212	0.992529
ViT (Standalone)	0.893266	0.807281	0.916334	0.871781
U-Net (Standalone)	0.487096	0.324141	0.845365	0.345092

Table 1. Segmentation performance of the evaluated models using Dice coefficient, IoU, precision, and recall.

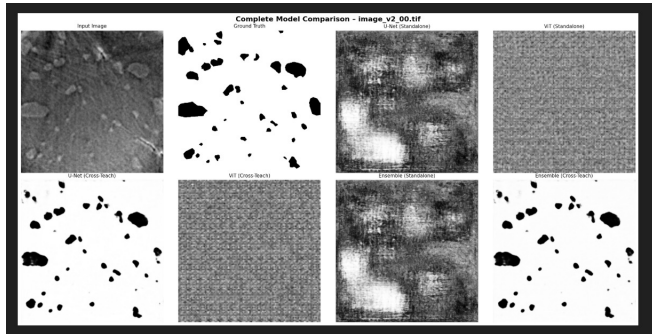


Fig. 1. Qualitative segmentation results from the cross-teaching U-Net, ViT, and ensemble models.

Figure 1 shows a visual of our cross-teaching segmentation. The stand-alone U-Net produces incomplete and irregular pore boundaries, which is consistent with its lower quantitative performance. The stand-alone ViT shows block-like artefacts and inconsistent pore shapes, despite perform-

ing better numerically. In contrast, the cross-teaching U-Net yields clear and well-defined pore masks, capturing both small and large structures accurately. The cross-teaching ViT also produces coherent segmentations that closely resemble the ground truth, aligning with its high Dice score. The ensemble output appears smooth and stable, combining the strengths of both models and producing pore boundaries that are visually consistent with the ground truth.

6. DISCUSSION

The results show that the cross-teaching framework improves the segmentation performance of both architectures, but the extent of improvement differs considerably. The U-Net benefits the most from the semi-supervised setup: its supervised baseline performs poorly due to the limited number of labelled slices, yet its performance increases substantially when pseudo-labels are introduced. This indicates that the additional supervision from unlabelled data is essential for stabilising its learning process and enabling it to capture fine pore structures. The ViT already performs reasonably well in the supervised setting and therefore gains a smaller absolute improvement by the cross-teaching method, but this variant achieves the highest overall accuracy. This suggests that transformer-based models can effectively incorporate pseudo-labels, even when trained with limited annotations. These differences can be linked to the architectural characteristics of the models. The U-Net’s convolutional filters require sufficient labelled examples to learn local patterns, whereas the ViT’s global attention mechanisms allow it to extract structural information earlier in its training. Further improvements could be achieved by increasing the number of labelled slices, refining confidence thresholds for pseudo-labels, or gradually introducing unlabelled data to reduce noise. Overall, the findings demonstrate that cross-teaching is a useful strategy for overcoming data scarcity, with the U-Net showing the strongest relative gain.

7. CONCLUSION

The study shows that semi-supervised cross-teaching is an effective strategy for improving pore segmentation performance in X-ray micro-tomography when supervised data is limited. Both the U-Net and ViT model both benefit from the use of unlabelled slices, with cross-teaching ViT achieving the strongest overall results. These findings highlight the potential of transformer-based architectures when supported by pseudo-labels and also confirming the reliability of U-net for this type of tasks as was found in literature. Although further labelled data would enhance the stability, the current results already show that high-quality segmentation can be achieved under limited annotated data. Overall the cross-teaching framework offers a promising direction for segmentation workflows in micro-tomography analysis.

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DECLARATION OF USE OF GENERATIVE AI

This declaration **must** be filled out and included as the **final page** of the document. The questions apply to all parts of the work, including research, project writing, and coding.

- We have used generative AI tools: Yes

If you answered *yes*, please complete the following sections.
List the generative AI tools you have used:

- ChatGPT, Claude AI, Google Gemini

Describe how the tools were used:

What did you use the tool(s) for?

ChatGPT was used to find sources for the related work and convert these sources. ChatGPT was also used to convert word text to latex and do a grammar check on the self typed text and adjust it if needed.

Claude was used to find relevant libraries and format/structure code. This was useful to organize/reformat old code and simplify functions/transitions.

Gemini was used to adapt old code into new scripts, especially for large files like notebooks with a lot of extra code/markdown/metadata. This included finding and extracting functions from earlier code versions.

At what stage(s) of the process did you use the tool(s)?

ChatGPT was used in the researching and writing phase.

Claude was used in the exploratory and creation phase.

Gemini was used in the final parts (to transfer code from complicated scripts to better-structured/smaller scripts).

How did you use or incorporate the generated output?

The output was used to read these papers and judge if they are applicable for information to write the related work section.

The output has been used to directly put in Overleaf

Most code output was used directly since it was just a reformatted (and commented) version of our own code.